INCH-POUND
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DETAIL SPECIFICATION

CLOTH, WIND RESISTANT SATEEN, COTTON AND NYLON

This specification is approved for use by all Departments and Agencies of the Department of Defense (DoD).

1. SCOPE

1.1 Scope. This specification covers wind resistant sateen cloth made of cotton and nylon blend.

1.2 Classification. This cloth is furnished in the following classes as specified.

1.2.1 Classes

Class 1 – Dyed
Class 2 – Dyed, Water Repellent Treated
Class 3 – Woodland Camouflage Pattern, Water Repellent Treated
Class 4 – Desert Camouflage Pattern (6 color), Water Repellent Treated
Class 5 – Desert Camouflage Pattern (3 color), Water Repellent Treated
Class 6 – Universal Camouflage Pattern, Water Repellent Treated
Class 7 – Operational Camouflage Pattern, Water Repellent Treated

Comments, suggestions, or questions on this document should be addressed: Attn: DLA Troop Support, 700 Robbins Avenue, Philadelphia, PA 19111-5096. Since contact information can change, you may want to verify the currency of this address information using Acquisition Streamlining and Standardization Information System (ASSIST) online database at https://assist.dla.mil.

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2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4, of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation or contract.

DRAWINGS

U.S. ARMY NATICK SOLDIER RESEARCH, DEVELOPMENT AND ENGINEERING CENTER

2-1-1516 – Woodland Camouflage Pattern
2-1-1884 – 6 Color Desert Pattern
2-1-2240 – 3 Color Desert Pattern
2-1-2519 – Universal Camouflage Pattern (UCP)
2-1-2592 - Operational Camouflage Pattern (OCP)

(Copies of drawings are available from the U.S. Army Natick Soldier Research, Development and Engineering Center, ATTN: RDNS-SES-WC, 10 General Greene Avenue, Natick, MA 01760-5019.)

FEDERAL ACQUISITON REGULATIONS (FAR)

52.209-4 First article Approval – Government Testing

(Copies are available online at https://www.acquisition.gov/browserfar.)

FEDERAL TRADE COMMISSION

Rules and Regulations under the Textile Fiber Products Identification Act

(Copies are available online at http://www.ftc.gov.)

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2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN ASSOCIATION OF TEXTILE CHEMISTS AND COLORISTS (AATCC)

AATCC Evaluation Procedure 1, Gray Scale for Color Change
AATCC Evaluation Procedure 2, Gray Scale for Staining
AATCC Evaluation Procedure 8, AATCC 9-Step Chromatic Transference Scale
AATCC Evaluation Procedure 9, Visual Assessment of Color Difference of Textiles
AATCC Test Method 8 - Colorfastness to Crocking: AATCC Crockmeter Method
AATCC Test Method 15 - Colorfastness to Perspiration
AATCC Test Method 16.3 - Colorfastness to Light: Xenon Arc
AATCC Test Method 20 - Fiber Analysis: Qualitative
AATCC Test Method 20A - Fiber Analysis: Quantitative
AATCC Test Method 22 - Water Repellency – Spray Test
AATCC Test Method 61 - Colorfastness to Laundering, Home and Commercial: Accelerated
AATCC Test Method 70 - Water Repellency: Tumble Jar Dynamic Absorption Test
AATCC Test Method 81 - pH of the Water Extract from Wet Processing Textiles
AATCC Test Method 118 - Oil Repellency: Hydrocarbon Resistance Test
AATCC Test Method 127 - Water Resistance: Hydrostatic Pressure Test
AATCC Test Method 135 - Dimensional Changes of Fabrics after Home Laundering

(Copies are available on line at www.aatcc.org.)

AMERICAN SOCIETY FOR QUALITY

ANSI/ASQ Z1.4 - Sampling Procedures and Tables for Inspection of Attributes

(Copies are available online at http://www.asq.org.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D737 - Standard Test Method for Air Permeability of Textile Fabrics
ASTM D1424 - Standard Test Method for Tear Resistance of Woven Fabrics by Falling-Pendulum (Elmendorf-Type) Apparatus
ASTM D1907/D1907M - Standard Test Method for Linear Density of Yarn (Yarn Number) by the Skein Method
ASTM D3775 - Standard Test Method for Fabric Count of Woven Fabric
ASTM D3776/D3776M - Standard Test Method for Mass per Unit Area (Weight) of Fabric
ASTM D3990 - Standard Terminology Relating to fabric Defects
ASTM D5034 - Standard Test Method for Breaking Force and Elongation of Textile Fabrics (Grab Test)
ASTM D5430 - Standard Test Method for Visually Inspecting and Grading Fabrics

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(Copies are available on line at www.astm.org.)

OTHER PUBLICATIONS

SDL ATLAS

Fabric Defect Replica Scales are available online at http://www.sdlatlas.com.) (See 6.5)

INFORMA HEALTHCARE

Repeat Insult Patch Test - Modified Draize Procedure –
Principles and Methods of Toxicology, A Wallace Hayes (editor).

(Copies are available online at https://www.crcpress.com.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.2.

3.2 Standard sample. The finished cloth shall match the standard sample for shade, appearance and pattern execution and shall, unless otherwise indicated, be equal to or better than the standard sample with respect to all characteristics for which the standard sample is referenced (see 6.3). The cloth shall be dyed, or dyed and printed on the filling effect side as the face (back side of the cloth).

3.3 Recycled, recovered, or environmentally preferable, or biobased materials. Recycled, recovered, or environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.3.1 Cotton. The cotton shall be carded.

3.3.2 Nylon. The nylon shall be first quality, high tenacity, semi-dull staple having a nominal cut length of 1-1/2-inches and a nominal denier of 2.25 to 2.5. The use of any form of nylon waste is prohibited. Testing shall be as specified in 4.4.1.

3.3.3 Yarn. The warp and filling yarn shall be singles made from a blend of 50 (± 5) percent nylon, with the remaining percentage cotton based on the dry weight of the desized cloth when tested as specified in 4.4.1.

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3.4 Color.

3.4.1 Class 1 and 2, dyed. The color of the finished cloth shall be a good match to Olive Green 107 or Camouflage Green 483 or as specified in the contract.

3.4.1.1 Class 3, Woodland Camouflage Pattern. The cloth shall be either dyed to a ground shade matching Light Green 354 and then overprinting with the remaining three (3) colors for the Dark Green 355, Brown 356, and Black 357 areas of the camouflage pattern or when the ground shade is not dyed to match Light Green 354, all four (4) colors of the camouflage pattern shall be printed to match all four (4) colors of the pattern.

3.4.1.2 Class 4, Desert Camouflage Pattern (6 Color). The cloth shall be either dyed to a ground shade matching Light Tan 379 and then overprinted with the remaining five (5) colors for Tan 380, Light Brown 381, Dark Brown 382, Black 383, and Khaki 384 areas of the camouflage pattern or when the ground shade is not dyed to Light Tan 379, all six (6) colors of the camouflage pattern shall be printed to match all six (6) colors of the pattern.

3.4.1.3 Class 5, Desert Camouflage Pattern (3 Color). The finished cloth shall be either dyed to a ground shade matching Light Tan 492 and then overprinted with the remaining two (2) colors for the Light Brown 493 and Light Khaki 494 areas of the camouflage pattern or when the ground shade is not dyed to match Light Tan 492, then all three (3) colors of the camouflage pattern shall be printed to match all three (3) colors of the pattern.

3.4.1.4 Class 6, Universal Camouflage Pattern (UCP). The finished cloth shall be either dyed to a ground shade matching Desert Sand 500 and then overprinted with the remaining two (2) colors for the Urban Gray 501 and Foliage Green 502 areas of the camouflage pattern or when the ground shade is not dyed to Desert Sand 500, all three (3) colors of the camouflage pattern shall be printed to match all three (3) colors of the pattern.

3.4.1.5 Class 7, Operational Camouflage Pattern (OCP). The finished cloth shall be either dyed to a ground shade matching Dark Cream 559 and then overprinted with the remaining six (6) colors Tan 525, Light Sage 560, Olive 527, Dark Green 528, Brown 529, and Bark Brown 561 areas of the camouflage pattern or when the ground shade is not dyed to Dark Cream 559 all seven (7) colors of the camouflage pattern shall be printed to match all seven (7) colors of the pattern.

3.4.2 Labile sulfur. Dyes and compounds containing elementary sulfur capable of oxidation to sulfuric acid shall not be used. The dyed cloth shall be “free” of labile sulfur when tested as specified in 4.4.1.

3.4.3 Visual shade matching (all classes). The color and appearance of the dyed and camouflage printed cloth shall match the standard sample when tested as specified in 4.4.1.

3.4.4 Colorfastness (all classes). The finished cloth shall conform to the colorfastness requirements listed below in Table I when tested as specified in 4.4.1 and 4.5.3.

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TABLE I. Colorfastness requirements (all classes).

Colors Evaluation	Laundering (3 cycles) (min.)	Light (40 AFU or 170 kJ/(m^{2}nm)@420 nm) 1/, (min.)	Perspiration (acid and alkaline) (min.)	Crocking (wet and dry) (min.)
All colors	--	--	3-4	--
All except Black 357 and Black 383	3-4	--	--	3.5
Black 357 and Black 383	3.0	--	--	1.0
Olive Green 107, Camouflage Green 483, Dark Green 355, Brown 356, Black 357 and 383, Dark Green 528, Brown 529, Dark Brown 382 and Bark Brown 561	--	3-4	--	--
Light Green 354, Light Tan 379 and 492, Tan 380, Light Brown 381 and 493, Khaki 384, Light Khaki 494, Desert Sand 500, Urban Gray 501, Foliage Green 502, Tan 525, Olive 527,Dark Cream 559 and Light Sage 560	--	3.0	--	--

1/AFU- AATCC Fading Units

3.5 Pattern execution (all classes except 1 and 2). The pattern on the printed finished cloth(s) shall match the standard sample in respect to design, colors and registration of the respective areas. The various areas of the pattern shall be properly registered in relation to each other and shall present definite sharp demarcations with a minimum of feathering or spew. Each pattern area shall show solid coverage; skitteriness, feathering, haloing or trapping and off register exceeding that shown on the standard sample in any of the printed areas will not be acceptable. When the standard sample is not available for pattern execution, a pattern drawing will be provided (see 6.3) and the pattern on the finished cloth shall match that of the drawing (see 2.2.2 and 6.3) when tested as specified in 4.4.1.
The pattern repeat for each class shall be as follows:

Class 3 Woodland Camouflage Pattern - 27.25 (+1.25, -2.50)-inches in the warp direction.
Class 4 and 5 Desert Camouflage Pattern three (3) and six (6) colors - 16.75 (+1.25, -1.75) inches in the warp direction.
Class 6 Universal Camouflage Pattern - 36.00 (+1.25, –2.50) inches in the warp direction.
Class 7 Operational Camouflage Pattern - 25.255 (+1.25, -2.50) inches in the warp direction and 68-inches maximum camouflage printed area in the filling direction.

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3.6 Spectral reflectance.

3.6.1 Class 1 and 2, dyed. The reflectance values shall conform to the requirements listed below, in Table II, when tested as specified in 4.4.1.

TABLE II. Spectral reflectance requirements for Class 1 and 2.

Wavelength, Nanometers (nm)	Olive Green 107		Camouflage Green 483	
	Min.	Max.	Min.	Max.
600	-	9	6	10
620	-	9	6	10
640	-	9	6	10
660	-	9	7	11
680	-	10	9	15
700	-	13	13	22
720	8	-	19	30
740	13	-	26	39
760	20	-	34	49
780	27	-	42	57
800	35	-	47	65
820	42	-	48	70
840	50	-	48	75
860	50	-	48	78

3.6.2 Class 3, Woodland Camouflage Pattern. The reflectance values shall conform to the requirements listed below, in Table III, when tested as specified in 4.4.1.

TABLE III. Spectral reflectance requirements for Class 3.

Wavelength, Nanometers (nm)	Light Green 354		Dark Green 355 and Brown 356		Black 357	
	Min.	Max.	Min.	Max.	Min.	Max.
600	8	18	3	9	-	10
620	8	18	3	9	-	10
640	8	18	3	9	-	10
660	8	18	3	12	-	10
680	10	22	3	14	-	10

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TABLE III. Spectral reflectance requirements for Class 3.- Continued

700	18	33	5	18	-	10
720	22	45	7	20	-	10
740	30	55	12	28	-	10
760	35	65	18	36	-	10
780	40	75	26	44	-	10
800	45	80	34	52	-	10
820	50	86	42	60	-	10
840	55	88	50	68	-	10
860	60	90	56	74	-	10

3.6.3 Class 4, Desert Camouflage Pattern (6 color). The reflectance values shall conform to the requirements listed below, in Table IV, when tested as specified in 4.4.1.

TABLE IV. Spectral reflectance requirements for Class 4.

Wavelength, Nanometers (nm)	Light Tan 379		Tan 380 and Khaki 384		Light Brown 381 and Dark Brown 382	
	Min.	Max.	Min.	Max.	Min.	Max.
700	38	53	25	44	19	41
720	38	54	25	45	20	41
740	39	55	25	46	20	42
760	40	56	26	47	21	42
780	41	57	27	48	21	42
800	43	58	28	50	22	43
820	45	59	30	52	23	45
840	48	62	33	55	24	46
860	50	65	36	58	25	48

3.6.3 Class 5, Desert Camouflage Pattern (3 color). The reflectance values shall conform to the requirements listed below, in Table V, when tested as specified in 4.4.1.

TABLE V. Spectral reflectance requirements for Class 5.

Wavelength, Nanometers (nm)	Light Tan 492		Light Brown 493		Light Khaki 494	
	Min.	Max.	Min.	Max.	Min.	Max.
700	38	53	19	41	25	44
720	38	54	20	41	25	45
740	39	55	20	42	25	46

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TABLE V. Spectral reflectance requirements for Class 5. - Continued

760	40	56	21	42	26	47
780	41	57	21	42	27	48
800	43	58	22	43	28	50
820	45	59	23	45	30	52
840	48	62	24	46	33	55
860	50	65	25	48	36	58

3.6.5 Class 6, Universal Camouflage Pattern (UCP). The reflectance values shall conform to the requirements listed below, in Table VI, when tested as specified in 4.4.1.

TABLE VI. Spectral reflectance requirements for Class 6.

Wavelength, Nanometers (nm)	Desert Sand 500		Urban Gray 501		Foliage Green 502	
	Min.	Max.	Min.	Max.	Min.	Max.
600	28	40	12	26	8	18
620	30	42	14	26	8	18
640	34	48	14	28	8	20
660	38	56	14	30	10	26
680	44	60	18	34	10	26
700	46	66	24	38	12	28
720	48	68	26	42	16	30
740	48	72	30	46	16	30
760	50	74	32	48	18	32
780	54	76	34	48	18	34
800	54	76	34	50	20	36
820	54	76	36	54	22	38
840	56	78	38	54	24	40
860	56	78	40	56	26	42

3.6.6 Class 7, Operational Camouflage Pattern (OCP). The reflectance values shall conform to the requirements listed below, in Table VII, when tested as specified in 4.4.1.

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TABLE VII. Spectral reflectance requirements for Class 7.

Wavelength, Nanometers (nm)	Dark Cream 559 & Tan 525		Light Sage 560, Olive 527 & Brown 529		Dark Green 528 & Bark Brown 561	
	Min.	Max.	Min.	Max.	Min.	Max.
600	22	44	12	30	3	11
620	24	45	12	30	3	12
640	24	45	12	32	4	13
660	25	45	12	32	4	14
680	28	45	14	34	4	17
700	28	48	14	36	6	23
720	30	52	16	39	6	23
740	32	55	18	41	10	25
760	36	56	20	43	14	30
780	38	57	22	45	18	35
800	40	57	22	45	21	40
820	44	58	24	46	24	42
840	46	59	26	47	26	43
860	48	60	28	48	28	45

3.7 Physical requirements. The finished cloth shall conform to the requirements listed below, in Table VIII, when tested as specified in 4.4.1.

TABLE VIII. Physical requirements.

Characteristic	All Classes
Weight, oz./sq.yd. (minimum)	8.5
Yarns per inch, (minimum) Warp Filling	 126 72
Breaking strength (pounds), (minimum) Warp Filling	 225 225
Tearing strength (pounds), (minimum) Warp Filling	 9.0 9.0
Air permeability, (cu.ft./min./sq.ft.) (maximum)	7.0

3.7.1 Weave. The weave when viewed from the face side shall be a 5-harness sateen. Testing shall be as specified in 4.4.1.

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3.7.2 Width. For government procurements only, the width of the finished cloth shall be as specified (see 6.2), and shall be the minimum acceptable width, inclusive of selvages.

3.8 Finish. The cloth shall be dyed, or dyed and printed on the filling effect side as the face (back side of the cloth). Cloths Classes 2, 3, 4, 5, 6 and 7 shall be given a durable water repellent treatment as specified in 3.8.1.

3.8.1 Water repellency (Classes 2, 3, 4, 5, 6 and 7). Classes 2, 3, 4, 5, 6 and 7 shall be given a durable water repellent treatment. Testing shall be as specified in 4.4.1.

TABLE IX. Water repellency (Classes 2, 3, 4, 5, 6 and 7).

	Dynamic absorption (percent) (Maximum)	Hydrostatic pressure (centimeter) (Minimum)	Spray rating 3/	Resistance to organic liquid (Minimum)
Initial	20 1/	30 2/	90, 90, 80	No wetting by n-tetradecane
After three (3) launderings	--	30 2/	--	--
After 15 launderings	20 1/	--	--	No wetting by n-tetradecane

1/ No individual specimen shall exceed 25 percent
2/ No individual specimen shall fall below 25 centimeters
3/ The results of the three individual determinations.

3.8.2 pH. The pH value of the water extract of the finished cloth shall be no less than 5.0 and no more than 8.5 when tested as specified in 4.4.1.

3.9 Dimensional stability. The shrinkage or elongation both in the warp and in the filling of the finished cloth shall not be greater than 2.5 percent for the individual sample unit and not greater than 2.0 percent for the lot average when tested as specified in 4.4.1.

3.10 Face identification. The Class 1 and 2 cloths filling side shall be identified as the face side by stamping that side with the word "Face" at each end of the roll.

3.11 Toxicity. The finished cloth shall not present a health hazard and shall show compatibility with prolonged, direct skin contact when tested as specified in 4.5.5. Chemicals recognized by the Environmental Protection Agency (EPA) as human carcinogens shall not be used.

3.12 Length and put-up. For Government procurements only, unless otherwise specified (see 6.2), the cloth shall be furnished in continuous lengths, each not less than 50 yards. Each length shall be put-up full width on a roll as specified in 5.1.

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3.13 Fiber identification. Each roll of the finished cloth shall be labeled or ticketed for fiber content in accordance with the Rules and Regulations under the Textile Fiber Products Identification Act.

3.14 Workmanship. The finished cloth shall conform to the quality of product established on this specification. The occurrence of defects shall not exceed the quality acceptance levels as specified in the contract or purchase order.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).
b. Conformance inspection (see 4.3).

4.2 First article inspection. A first article, submitted in accordance with 3.1, shall be inspected, visually examined for appearance, color, pattern execution, finished defects in Table X and tested for the characteristics as specified in Table XI.

4.3 Conformance inspection. Conformance inspection shall include the visual examination of 4.4, and the tests of 4.4.1 through 4.5.5 as applicable. Sampling for inspection shall be performed in accordance with ANSI/ASQ Z1.4 and with quality acceptance levels (AQL) as specified in the contract and/or order, except where otherwise indicated.

4.3.1 Material inspection. As specified in 4.1, the material shall be inspected in accordance with all the requirements of referenced documents, unless otherwise excluded, amended, modified or qualified in this specification or applicable procurement documents.

4.4 Yard by yard examination. Each roll in the sample shall be examined yard-by-yard (full width excluding selvage) on the face side only with both overhead and back lighting. The defects are as defined in ASTM D3990 and only defects that are clearly noticeable at normal inspection distance (3-feet) shall be scored. The defects are assigned demerit points in accordance with ASTM D 5430, Option A, with the exceptions listed below:

a. The defects found shall be counted regardless of their proximity to each other, except where two (2) or more defects represent a single local condition of the cloth, in which case only the more serious defect shall be counted.

b. Only coarse yarns that exceed twice or more the normal diameter shall be scored.

c. Mixed filling shall be scored only if attributed to shade bar resulting from wrong ply, wrong twist in the yarn or off shade yarn.

d. The defects specified in Table X shall be scored four (4) points for each yard in which they occur.

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TABLE X. - Defects that score four (4) points.

Defect Description
Any hole, cut, tear or abrasion mark.
Edge ravels when pulled outward.
Uneven weaving throughout.
Broken or missing yarn.
Any spot, stain or streak more than 1 inch in combined directions.
Knots greater than Fabric Defect Replica Scale Level D (see 2.3).
Slubs greater than Fabric Defect Replica Scale Level F (see 2.3).
Area of poor dye penetration, streak or shade bar.
Width not as specified within a tolerance of no less than 1/2-inch.
Objectionable odor.
Slack or tight selvages.
Any embedded foreign matter or uncleanness.
Edges rolled folded, scalloped or corded, baggy, ridgy or wavy cloth.

4.4.1 End item testing. The cloth shall be tested for the characteristics listed in Table XI. The methods of testing as specified wherever applicable and as listed in Table XI shall be followed. All test reports shall contain the individual values utilized in expressing the final results. The sample unit shall be five (5) continuous yards full width of the finished cloth for all physical and chemical tests. The lot shall be unacceptable if one test fails to meet any requirement specified.

The sample size shall be in accordance with the following:

Lot Size (yards)	Sample size (sample units)
800 or less	2
801 up to and including 22,000	3
22,001 and over	5

TABLE XI. End item tests.

Characteristic	Requirement paragraph	Test method
Cotton identification	3.3.1	AATCC 20 (see 6.4)
Nylon identification	3.3.2	AATCC 20 (see 6.4)
Denier, nylon	3.3.2	ASTM D1907/D1907M
Fiber content (cotton and nylon)	3.3.3	AATCC 20A (see 6.4)
Yarn ply	3.3.3	Visual
Presence of labile sulfur	3.4.2	4.5.1

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TABLE XI. End item tests. - Continued

Visual shade matching	3.4.3	4.5.2
Colorfastness:		4.5.3
Light	3.4.4	AATCC 16.3, Option 3 1/
Laundering	3.4.4	AATCC 61, Test 3A 2/, 3/
Perspiration	34.4	AATCC 8 2/, 3/
Crocking	3.4.4	AATCC 15 4/
Pattern execution	3.5	Visual and linear measurement using a precision steel ruler
Spectral reflectance:		
Class 1 and 2	3.6.1	4.5.4
Class 3	3.6.2	4.5.4
Class 4	3.6.3	4.5.4
Class 5	3.6.4	4.5.4
Class 6	3.6.5	4.5.4
Class 7	3.6.6	4.5.4
Weight	3.7	ASTM D3776/3776M
Breaking strength	3.7	ASTM D5034
Tearing strength	3.7	ASTM D1424
Yarns per inch	3.7	ASTM D3775
Air permeability	3.7	ASTM D737
Weave	3.7.1	Visual
Water repellency (Class 2, 3, 4, 5, 6 and 7):		
Dynamic absorption:	3.8.1	
Initial		AATCC 70
After 15 launderings		AATCC 135 (3)(V)(A)iii and AATCC 70
Hydrostatic pressure:	3.8.1	
Initial		AATCC 127
After three (3) launderings		AATCC 135 (3)(V)(A)iii and AATCC 127
Spray rating:	3.8.1	
Initial		AATCC 22
Resistance to organic liquids:	3.8.1	
Initial		AATCC 118
After 15 launderings		AATCC 135 (3)(V)(A)iii and AATCC 118
pH	3.8.2	AATCC 81
Dimensional stability after three (3) cycles	3.9	AATCC 135 (3)(V)(A)iii
Toxicity	3.11	4.5.5

1/ Rated using the AATCC Evaluation Procedure 1, Gray Scale for Color
2/ Rated using the AATCC Evaluation Procedure 1, Gray Scale for Color Change and AATCC Evaluation Procedure 2, Gray Scale for Staining.
3/ Only the stain on the nylon and cotton fibers of the color transfer cloth shall be evaluated.

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4/ Rated using the AATCC Evaluation Procedure 8, AATCC 9-step Chromatic Transference Scale.

4.5 Methods of testing or inspection.

4.5.1 Presence of labile sulfur test. In the determination of presence of labile sulfur in textile materials with lead acetate, two (2) 1.50 (±) 0.01 gram samples from each material submitted for evaluation shall be tested. Each of the two (2) samples shall be cut into very small pieces and placed into separate test tubes. The samples shall submersed in a stannous chloride solution that contains 100 grams of stannous chloride crystals ACS grade in a 100 milliliters of hydrochloric acid ACS (35 percent concentration) and 50 milliliters of distilled water. A filter paper wet out with a 5.0 percent lead acetate solution shall be placed over the top of the test tube. The lead acetate solution contains 5.0 grams of lead acetate CP reagent grade and enough distilled water to make up a 100 milliliter solution; if the solution is not clear, add a few drops (one at a time) of glacial acetic acid until the solution is clear. The test tube containing the textile sample, stannous chloride and wet filter paper shall be heated over a low flame until the solution is boiling. The solution should not be heated for more than 15 seconds. A brown to black stain on the filter paper should be evaluated as follows:

Free - The filter paper shows no discoloration or staining of any kind.
Slight - The filter paper shows a light tan to light brown discoloration stain.
Moderate - The filter paper shows a dark brown discoloration stain.
Severe - The filter paper shows a black color stain.

4.5.2 Visual shade matching evaluation. The color and appearance of the cloth shall match the standard sample when viewed using the AATCC Evaluation Procedure 9, Option A, with sources simulating artificial daylight D75 illuminant with a color temperature of 7500 (± 200) K illumination of 100 (± 20) foot candles, and shall be a good match to the standard sample under incandescent A illuminant with a color temperature of 2856 (± 200) K.

4.5.3 Colorfastness evaluation testing. When testing for colorfastness properties, each color shall be evaluated, whenever possible, separately and reported as such. In cases where the print pattern does not allow for the evaluation of each color separately, the test results should indicated which colors were evaluated together.

4.5.3.1 Operational Camouflage Pattern (OCP) (Class 7). All colorfastness testing for OCP (Class 7) shall be performed on the solid color area and not the tonal area.

4.5.4 Spectral reflectance test. Spectral reflectance data shall be obtained from 600 to 860 nanometer (nm) for Class 1, 2, 3, 6 and 7 and 700 nm to 860 nm for Class 4 and 5 at 20nm intervals on a spectrophotometer relative to the polytetrafluoroethylene (PTFE) family of compounds, the preferred white standard. Other white reference materials may be used provided they are calibrated to absolute white or vitrolite tiles. The spectral band width shall be less than 20 nm at 860 nm. Reflectance measurements shall be made by either the monochromatic or polychromatic mode of operation. When the polychromatic mode of operation is used, the spectrophotometer shall operate with the specimen diffusely illuminated with the full emission of

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a continuous source that simulates in the visible spectrum either CIE Source A or CIE Source D65. The specimen shall be viewed at an angle of no greater than 10º from normal, the specular component included. Measurements shall be taken on a minimum of two (2) different areas and the data averaged. The Woodland Camouflage Pattern (Class 3) cloth shall be measured as a single layer using three (3) backing layers of the same shade for Light Green 354, Dark Green 355 and Brown 356 colors and two (2) backing layers of the same shade for Black 357. The Desert (6 and 3-color) patterns (Class 4 and 5), Universal Camouflage Pattern (Class 6) and Operational Camouflage Pattern (OCP) (Class 7) shall be measured as a single layer backed with four (4) layers of the same shade cut from the standard. The spectral reflectance on OCP shall be measured on the mostly solid color area not the tonal area. Specimens shall be oriented in different directions during testing. When possible, the specimens tested shall not contain the same warp and filling yarns. The diameter for standard aperture size used in the color measurement device shall be 1.0 to 1.25-inches for Classes 3, 4 and 5, and 0.3725-inches or larger for the Classes 6 and 7. The measured areas should be at least 6-inches away from the selvage. Photometric accuracy of the spectrophotometer shall be within one (1) percent and wavelength accuracy within two (2) nm. Any color having spectral reflectance values falling outside the limits at four (4) or more of the wavelengths specified shall be considered a test failure.

4.5.5 Toxicity test. When required, (see 6.2) an acute dermal irritation study and a skin sensitization study shall be conducted. When the results of these studies indicate the material is not a sensitizer or irritant, a Repeat Insult Patch Test shall be performed in accordance with the Modified Draize Procedure (see 2.3). If toxicity requirement (see 3.12) can be demonstrated with historical use data, toxicity testing may not be required (see 6.2).

4.5.5.1 Toxicity documents. All finishes/chemicals used to process the garment shall be identified and accompanied by the appropriate Material Data Sheet (MDS) information. The use of chemicals recognized by the Environmental Protection Agency (EPD) as human carcinogens is prohibited.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of material is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Department or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

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6.1 Intended use. The cloth is intended for use in clothing where a high degree of wind resistance and water resistance is of prime importance.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. Class of cloth required (see 1.2)
c. The specific issue of individual documents referenced (see 2.2)
d. When first article is required (see 3.1 and 4.2).
e. Length required if other than specified (see 3.12).
f. Conformance inspection acceptance quality limits (AQL) (see 4.3)
g. Inspection conditions (see 4.1)
h. When toxicity testing is required (see 3.11 and 4.5.5)
i. Packaging (see 5.1).

6.3 Standard sample. For access to samples and pattern drawings, address the contracting activity issuing the invitation for bids or request for proposal.

6.4 Certificate of compliance. The contracting activity may select to accept a certificate of compliance for stated requirement.

6.5 Fabric defect replica scales. Fabric Defect Replica Scales are available for purchase from SDL Atlas. Contact Customer Service 1-803-329-2110 and request item number is 402985 – Slub/Knot Replica Set.

6.6 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issues, due to the extensiveness of the changes.

6.7 Subject term (key word) listing.

Clothing
Desert
Dyed
Operation Camouflage Pattern (OCP)
Printed cloth
Universal (UCP)
Water repellent treated
Woodland

Custodians:
Army - GL
Navy - NU
Air Force - 11

Preparing activity:
DLA - CT

Project: 8305-2018-008

Reviewer activities
Army - MD

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